Obesity

What is Obesity?

- Obesity is a **chronic metabolic disorder** characterized by an excessive accumulation of body fat, leading to serious health complications.
- It is increasing rapidly in developing countries due to **lifestyle changes**

Etiology of Obesity

1. Energy Imbalance

•Occurs due to **excessive calorie intake** and **insufficient physical activity**.

2. Genetic Factors

•Obesity is **polygenic**, with multiple genes contributing to susceptibility when combined with **environmental triggers**.

3. Underlying Diseases

•Conditions such as **hypothyroidism** can slow metabolism and lead to weight gain.

4. Medication-Induced Obesity

•Some drugs are associated with **weight gain**, including:

•**Antidepressants** (tricyclic antidepressants, lithium)

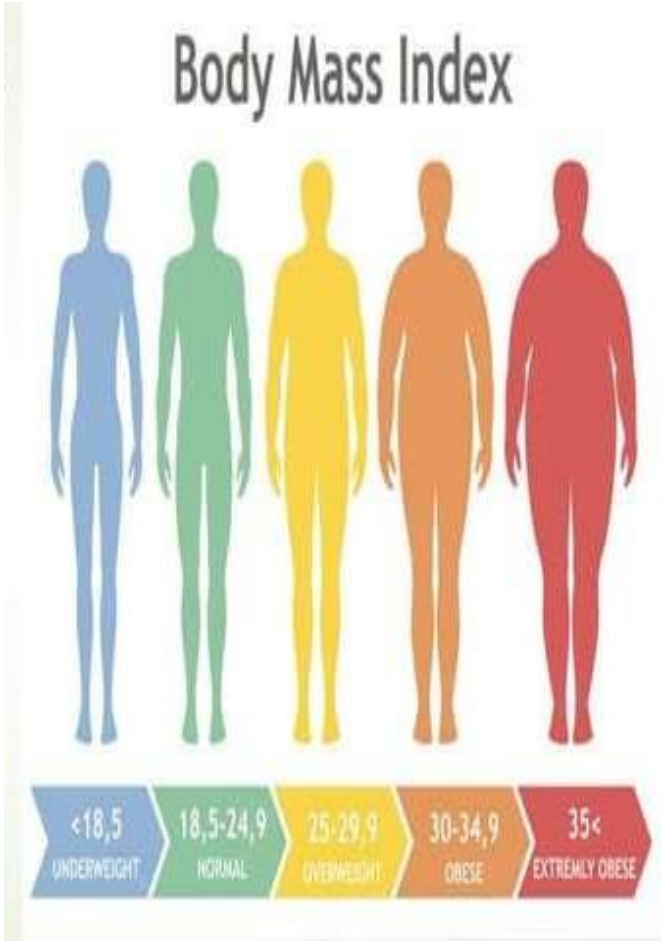
•**Antiepileptics** (valproate, carbamazepine)

•**Insulin and certain oral hypoglycemics**

Body Mass Index (BMI)

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BMI = Weight (kg) / Height² (m²)
A simple tool to classify **overweight and obesity** across all ages and genders.

BMI (kg/m^2)	**Classification**
< 18.5	Underweight
18.5 – 24.9	Normal weight
25 – 29.9	Overweight
≥ 30	Obesity



Health Consequences of Obesity

Obesity is a major risk factor for multiple chronic diseases, including:

a. Metabolic & Cardiovascular Diseases

- **Type 2 Diabetes Mellitus (T2DM)**: Insulin resistance caused by excessive adipose tissue can lead to **chronic hyperglycemia**.
- **Hypertension & Atherosclerosis**: Excess body weight contributes to increased blood pressure and cholesterol levels, raising the risk of **stroke and heart disease**.

b. Musculoskeletal Disorders

- **Osteoarthritis**: Increased mechanical stress on joints leads to **cartilage degradation and chronic pain**.
- **Lower Back Pain**:

Health Consequences of Obesity

c. Psychological & Social Impact

- **Depression & Anxiety**: Obesity is linked to **low self-esteem, emotional distress, and social isolation**.

- Managing obesity requires a combination of **diet, exercise, and, in some cases, pharmacological interventions, including herbal remedies**.

Why Consider Herbal Remedies for Weight Loss?

While **caloric restriction, physical activity, and behavioral modifications** remain the foundation of weight management, many individuals **struggle with long-term adherence** to lifestyle changes.

- Pharmaceutical weight-loss medications, such as **orlistat, phentermine, and GLP-1 receptor agonists (e.g., semaglutide)**, have been shown to be effective but often come with **side effects, high costs, and long-term safety concerns**.

Why Consider Herbal Remedies for Weight Loss?

- Recent research suggests that **combining herbal extracts with lifestyle interventions** may offer a **safer and more sustainable** approach to weight loss compared to pharmaceutical interventions alone.
- However, **standardization, quality control, and proper dosing** remain key challenges in the widespread adoption of herbal weight-loss treatments.

1. Dandelion (*Taraxacum officinale*)



Part Used:

- Dried leaves, dried roots, leaf juice, root tincture.

Active Constituents:

- Taraxacin, taraxasterol,
- Rich in **potassium, vitamins, and minerals**.

Mechanism of Action:

✓ **Diuretic effect** – Increases urine output, reducing water retention.
✓ **Laxative effect** – Promotes bowel movements.
✓ **Cholagogue activity** – Stimulates bile secretion, enhancing fat digestion and metabolism.

- ⚠ Not recommended for individuals with **kidney disease** due to its **high potassium content**.

2. Garcinia (*Garcinia gambogia*)



Part Used:

- Fruit rind (available in capsules, tablets, powders).

Active Constituents:

- **Hydroxycitric acid (HCA)** – The key bioactive compound.

Mechanism of Action:

- ✓ **Inhibits Citrate Lyase**, an enzyme that converts carbohydrates into stored fat. (reduce fatty acid synthesis and to reduce plasma cholesterol levels)
  ✓ **Suppresses appetite** by increasing serotonin levels in the brain. , and the immediate result of this blockage is that the liver is prompted to then release glycogen as a fuel source

3. Ginger (*Zingiber officinale*)

Part Used:

- Rhizome.

Active Constituents:

V. O containing Gingerols, shogaols, zingerone.

Mechanism of Action:

- ✓ **Boosts metabolism** – Can increase energy expenditure by up to **20%**.
  ✓ **Improves digestion**, enhancing nutrient absorption and gut health.

Weight Loss Detox Tea Recipe:

- 500 mL green tea
- Juice of 1 lemon
- 2–3 pieces of fresh ginger
- 3 teaspoons honey

4. Guggul (*Commiphora wightii*)



- **Part Used**:
- Resinous sap of bark (gum guggul, guggulipid).
- **Active Constituents**:
- Guggulsterones, guggulsterols, myrcene.
- **Mechanism of Action**:
  - **Lowers LDL and triglycerides** while increasing HDL.
  - Increases the elimination of cholesterol, while reducing fat absorption from intestine
  - **Enhances thyroid hormone production (T3, T4)**, which boosts metabolism.

5. Guarana (*Paullinia cupana*)



- **Part Used**:
- Seeds.
- **Active Constituents**:
- High caffeine content (2-5 times more than coffee beans)
- **Mechanism of Action**:
  - **Increases thermogenesis**, enhancing fat burning.
  - **Acts as a diuretic**, promoting water loss.
- **Precautions**:
  - Avoid combining with other stimulants (risk of **hypertension**).
  - Can interact with **anticoagulants (e.g., warfarin)**.
  - **Commercial Energy Drinks**: Popular brands (like Red Bull use guarana as a caffeine source.

5. Guarana (*Paullinia cupana*)

- • Based on its caffeine content, it is advised that doses should not exceed that amount that will provide approximately 250 mg of caffeine daily. This is equivalent to 2.5-4 g guarana/day.
- As YGD, which a combination of yerba mate, guarana and damiana significantly delayed gastric emptying, reduced the time to perceived gastric fullness and induced significant weight loss over 45 days in overweight patients.

6. Guar Gum (*Cyamopsis tetragonoloba*)



Part Used:

- Seeds (high fiber content).
- **Mechanism of Action**:
- Fibers of guar gum bind to fats and eliminate them with feces. This reduces the level of LDL cholesterol
- herb fibers the stomach and provides a feeling of fullness that promote satiety reducing calorie intake

7. Green Tea (*Camellia sinensis*)

- **Active Constituents: Epigallocatechin gallate (EGCG), caffeine.**
- **Mechanism of Action:**
  - **Increases metabolism.**
  - **Enhances fat oxidation.**
  - Effect similar to Guarana but less potent
- **Clinical Evidence:**
  - A **690 mg daily dose** of green tea extract for **3 months** resulted in significant weight loss.

Cocoa

- Cocoa also contributes to weight loss by:
- Inhibiting digestive enzymes such as lipases and amylases
- Modifying the expression of genes responsible for proteins involved in fatty acid transportation and thermogenesis; these effects result in lower body weight, fat accumulation and triglyceride levels
- ➞Increasing the release of adiponectin, a protein hormone produced in the adipose tissue and released into the blood. Adiponectin reduces the amount of fat stored in the body.



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